

1) Conversion

a) Decimal

Convert

Input a decimal number to format as single-precision representation.

DECIMAL NUMBER

123824

Convert

Accepts standard decimals, scientific notation (e), Infinity, -Infinity and NaN.

SIGN	EXPONENT	MANTISSA
0 1 bit	10001111 8 bits	11100011101100000000000 23 bits

BINARY 0 10001111 11100011101100000000000

HEX 0x47F1D800

b) Floating Point

Convert

Input a decimal number to format as single-precision representation.

DECIMAL NUMBER

6.75

Convert

Accepts standard decimals, scientific notation (e), Infinity, -Infinity and NaN.

SIGN	EXPONENT	MANTISSA
0 1 bit	10000001 8 bits	10110000000000000000000 23 bits

BINARY 0 10000001 10110000000000000000000

HEX 0x40D00000

c) Hexadecimal

The screenshot shows the IEEE 754 Computing Machine web application. The browser tabs include "New Tab" and "Convert - IEEE 754 Machine". The address bar shows "127.0.0.1:5000/convert". The page title is "IEEE 754 COMPUTING MACHINE" with a "Menu" link. A small display at the top shows "0x00". The main heading is "Convert" with the instruction "Input a decimal number to format as single-precision representation." Below this is a form with a "DECIMAL NUMBER" label, a text input field containing "DE2AD21", and a "Convert" button. A message below the input field states: "Accepts standard decimals, scientific notation (e), Infinity, -Infinity and NaN." A red error message box at the bottom of the form says: "'DE2AD21' is not a valid decimal number." At the very bottom of the page, there is a navigation bar with the text "sign · exponent · mantissa".

d) Scientific Notation

The screenshot shows the IEEE 754 Computing Machine web application with the input "923e4". The "Convert" button is active. Below the input field, the bit-level breakdown of the IEEE 754 single-precision representation is shown:

SIGN	EXPONENT	MANTISSA
0 1 bit	10010110 8 bits	00011001101011010110000 23 bits

Below the bit-level breakdown, the binary and hexadecimal representations are displayed:

BINARY: 0 10010110 00011001101011010110000

HEX: 0x4B0CD6B0

e) +/- Infinity

0x00

Convert

Input a decimal number to format as single-precision representation.

DECIMAL NUMBER

Infinity

Convert

Accepts standard decimals, scientific notation (e), Infinity, -Infinity and NaN.

SIGN

0

1 bit

EXPONENT

11111111

8 bits

MANTISSA

00000000000000000000000

23 bits

BINARY

0 11111111 00000000000000000000000

HEX

0x7F800000

0x00

Convert

Input a decimal number to format as single-precision representation.

DECIMAL NUMBER

-Infinity

Convert

Accepts standard decimals, scientific notation (e), Infinity, -Infinity and NaN.

SIGN

1

1 bit

EXPONENT

11111111

8 bits

MANTISSA

00000000000000000000000

23 bits

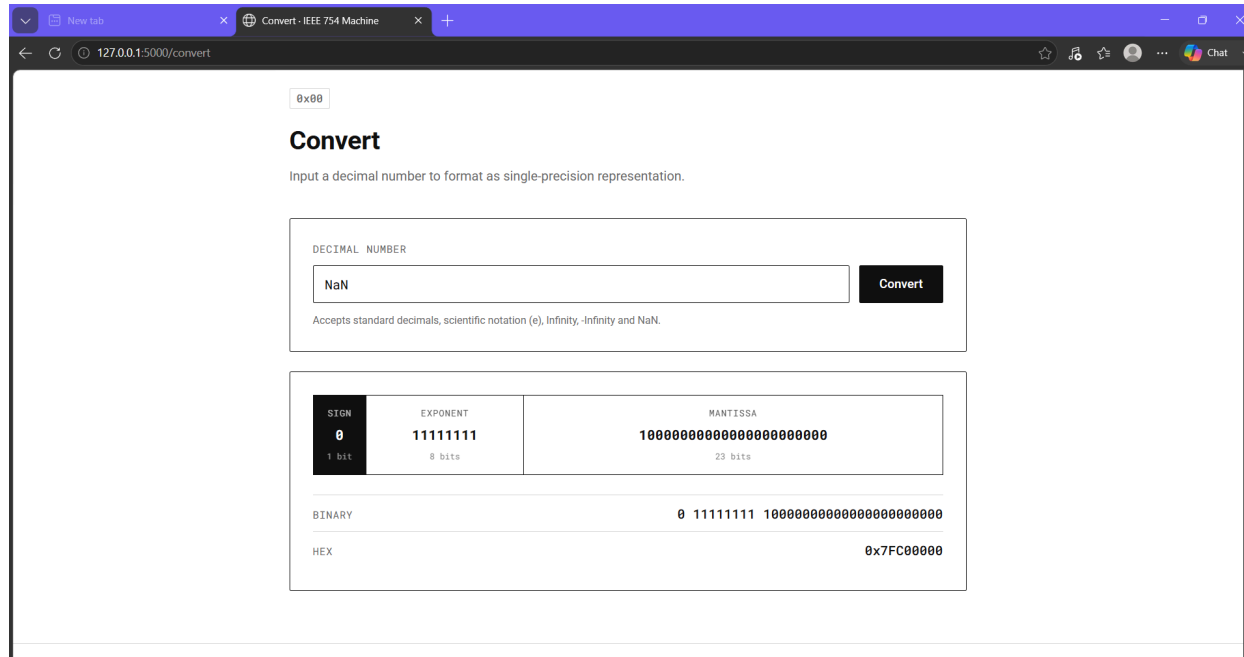
BINARY

1 11111111 00000000000000000000000

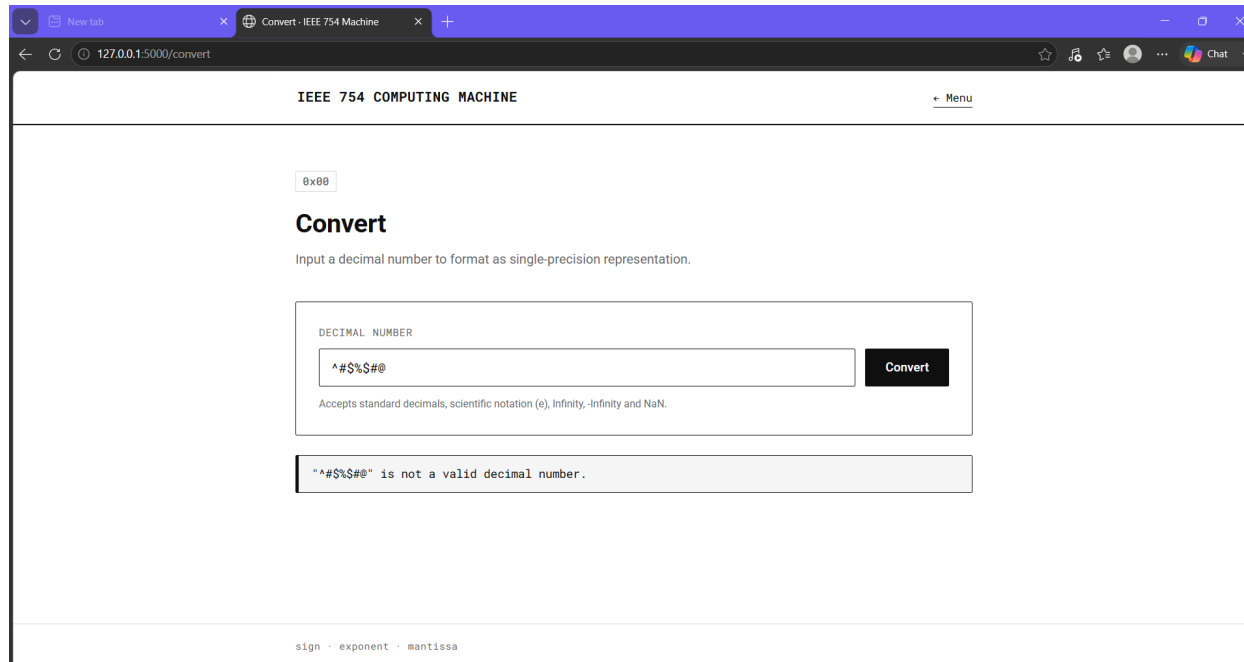
HEX

0xFF800000

f) NaN



g) Non-Digit Characters



2) Rounding

a) Decimal

The screenshot shows a web browser window with the URL `127.0.0.1:5000/rounding`. The page is titled "Rounding" and has the subtitle "One value, four ways to cut it off." Below the title, there is a form with the following fields:

- INPUT FORMAT:** Two radio buttons, "Decimal" (selected) and "Binary".
- DECIMAL NUMBER:** A text input field containing "12345".
- DIGITS AFTER THE DECIMAL POINT:** A text input field containing "2".
- Round:** A black button with white text.

Below the form, the result is displayed: "12345 rounded to 2 digit(s)". There are four boxes showing different rounding methods:

Truncated round toward zero	Rounded up round toward ∞	Rounded down round toward $-\infty$	Ties to even nearest, IEEE 754 default
12000	13000	12000	12000

b) Floating-point

The screenshot shows the same web browser window with the URL `127.0.0.1:5000/rounding`. The page is titled "Rounding" and has the subtitle "One value, four ways to cut it off." Below the title, there is a form with the following fields:

- INPUT FORMAT:** Two radio buttons, "Decimal" (selected) and "Binary".
- DECIMAL NUMBER:** A text input field containing "4.834".
- DIGITS AFTER THE DECIMAL POINT:** A text input field containing "5".
- Round:** A black button with white text.

Below the form, the result is displayed: "4.834 rounded to 5 digit(s)". There are four boxes showing different rounding methods:

Truncated round toward zero	Rounded up round toward ∞	Rounded down round toward $-\infty$	Ties to even nearest, IEEE 754 default
4.8340	4.8340	4.8340	4.8340

c) Binary

The screenshot shows a web browser window with the URL `127.0.0.1:5000/rounding`. The page title is "Rounding" and the subtitle is "Input a decimal or binary value to demonstrate four ways of rounding-off."

The interface has two main sections. The top section is titled "INPUT FORMAT" and contains two radio buttons: "Decimal" and "Binary". The "Binary" button is selected. Below this is a "BINARY NUMBER" input field containing the value `101010110`. Below that is a "SIGNIFICANT BITS" input field containing the value `2`. A small text note below the input fields says "Counted from the leading 1 – e.g. 1.1011 to 3 significant bits is 1.10." Below the input fields is a "Round" button.

The bottom section shows the result of the rounding operation. It says "101010110 rounded to 2 significant bit(s)". Below this is a table with four columns representing different rounding methods:

Truncated round toward zero	Rounded up round toward +∞	Rounded down round toward -∞	Ties to even nearest, IEEE 754 default
100000000	100000001	100000000	100000001

d) Numeric Type Mismatch

The screenshot shows the same web browser window with the URL `127.0.0.1:5000/rounding`. The page title is "Rounding" and the subtitle is "Input a decimal or binary value to demonstrate four ways of rounding-off."

The interface has two main sections. The top section is titled "INPUT FORMAT" and contains two radio buttons: "Decimal" and "Binary". The "Binary" button is selected. Below this is a "BINARY NUMBER" input field containing the value `492843ADE`. Below that is a "SIGNIFICANT BITS" input field containing the value `2`. A small text note below the input fields says "Counted from the leading 1 – e.g. 1.1011 to 3 significant bits is 1.10." Below the input fields is a "Round" button.

The bottom section shows an error message: `"492843ADE" is not a valid binary number (use only 0 and 1).`

0x01

Rounding

Input a decimal or binary value to demonstrate four ways of rounding-off.

INPUT FORMAT

☒ Decimal ☐ Binary

DECIMAL NUMBER

AED12512

SIGNIFICANT DIGITS

2

Counted from the first nonzero digit – e.g. 10.1234 to 3 significant digits is 10.1.

Round

"AED12512" is not a valid decimal number.

e) Non-digit Characters

0x01

Rounding

Input a decimal or binary value to demonstrate four ways of rounding-off.

INPUT FORMAT

☒ Decimal ☐ Binary

DECIMAL NUMBER

%@#%@^%#&

SIGNIFICANT DIGITS

2

Counted from the first nonzero digit – e.g. 10.1234 to 3 significant digits is 10.1.

Round

"%@#%@^%#&" is not a valid decimal number.

f) Negative Numbers

The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5000/rounding". The page title is "Rounding - IEEE 754 Machine". The interface is divided into several sections:

- INPUT FORMAT:** Two radio buttons are present: "Decimal" (selected) and "Binary".
- DECIMAL NUMBER:** A text input field containing the value "-123436".
- SIGNIFICANT DIGITS:** A text input field containing the value "2".
- Counted from the first nonzero digit – e.g. 10.1234 to 3 significant digits is 10.1.**
- Round:** A button to perform the rounding operation.
- Result:** A section titled "-123436 rounded to 2 significant digit(s)" containing a table with four options:

Truncated	Rounded up	Rounded down	Ties to even
round toward zero	round toward $+\infty$	round toward $-\infty$	nearest, IEEE 754 default
-120000	-120000	-130000	-120000

3) Arithmetic

a) Decimal -> Decimal (Addition)

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

Add or multiply two operands, bit by bit.

OPERAND A
☒ Decimal ☐ Hex
123213

OPERAND B
☒ Decimal ☐ Hex
31231

OPERATION
☒ Addition (A + B) ☐ Multiplication (A × B)
Compute

123213.0 + 31231.0 = 154444.0

SIGN 0 1 bit	EXPONENT 10010000 8 bits	MANTISSA 00101101101001100000000 23 bits
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BINARY 0 10010000 001011011010011000000000

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

BINARY 0 10010000 001011011010011000000000

HEX 0x4816D300

DECIMAL 154444.0

Step-by-step

- Convert operand A to single-precision format**
A = 123213.0 -> sign=0, significand=1.11100001010011010000000 x 2¹⁶
- Convert operand B to single-precision format**
B = 31231.0 -> sign=0, significand=1.1110011111111000000000 x 2¹⁴
- Align exponents**
Common exponent reference: 2¹⁴. A shifted left 2 bit(s), B shifted left 0 bit(s) to express both significands at the same scale.
- Perform addition**
Same sign -> add magnitudes: 11110000101001101000000000 + 111100111111110000000000.
- Normalize**
Raw result 10010110110100110000000000 -> leading 1 at 2¹⁷.
- Round (ties-to-even)**
Unrounded mantissa: 1.00101101101001100000000000 x 2¹⁷. Rounded to 23 bits: 001011011010011000000000.
- Assemble result**
Final 32-bit representation: 0x4816D300

b) Hex -> Hex (Addition)

Arithmetic - IEEE-754 Machine

127.0.0.1:5000/arithmetic

Chat

Arithmetic

Add or multiply two operands, bit by bit.

OPERAND A

☐ Decimal

☒ Hex

FACE

OPERAND B

☐ Decimal

☒ Hex

ECAF

OPERATION

☒ Addition (A + B)

☐ Multiplication (A × B)

Compute

8.99717692004392e-41 + 8.490607525190499e-41 = 1.748778444523442e-40

SIGN 0 1 bit	EXPONENT 00000000 8 bits	MANTISSA 000000111100111011111101 23 bits
BINARY		
0 00000000 000000111100111011111101		

HEX

0x0001E77D

DECIMAL

1.748778444523442e-40

Step-by-step

01 Convert operand A to single-precision format

A = 8.99717692004392e-41 -> sign=0, significand=0.00000001111101011001110 x 2⁻¹²⁶ (subnormal)

02 Convert operand B to single-precision format

B = 8.490607525190499e-41 -> sign=0, significand=0.000000011101001010111 x 2⁻¹²⁶ (subnormal)

03 Align exponents

Common exponent reference: 2⁻¹²⁶. A shifted left 0 bit(s), B shifted left 0 bit(s) to express both significands at the same scale.

04 Perform addition

Same sign -> add magnitudes: 1111101011001110 + 1110110010101111.

05 Normalize

Raw result 11110011101111101 -> leading 1 at 2⁻¹³³.

06 Round (ties-to-even)

Unrounded mantissa: 1.1110011101111101 x 2⁻¹³³. Rounded to 23 bits: 1110011101111010000000.

07 Subnormal result

Biased exponent -6 <= 0 -> result is subnormal (denormalized), shifted right by 6 bit(s).

c) Hex -> Hex (Multiplication)

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

Chat

Arithmetic

Add or multiply two operands, bit by bit.

OPERAND A

☐ Decimal
☒ Hex

FACE

OPERAND B

☐ Decimal
☒ Hex

ECAF

OPERATION

☐ Addition (A + B)
☒ Multiplication (A × B)

Compute

8.99717692004392e-41 × 8.490607525190499e-41 = 0.0

SIGN	EXPONENT	MANTISSA
0 1 bit	00000000 8 bits	00000000000000000000000 23 bits

BINARY0 00000000 000000000000000000000000

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

Chat

BINARY0 00000000 000000000000000000000000

HEX0x00000000

DECIMAL0.0

Step-by-step

01 Convert operand A to single-precision format

A = 8.99717692004392e-41 -> sign=0, significand=0.000000111101011001110 × 2⁻¹²⁶ (subnormal)

02 Convert operand B to single-precision format

B = 8.490607525190499e-41 -> sign=0, significand=0.00000011101100101111 × 2⁻¹²⁶ (subnormal)

03 Multiply significands, add exponents

Significands: 0000000111101011001110 × 000000011101100101111. Exponents: -126 + -126 = -252. Result sign = 0 XOR 0 = 0.

04 Normalize

Raw product 1110011111000010101101011010010 -> leading 1 at 2⁻²⁶⁷.

05 Round (ties-to-even)

Unrounded mantissa: 1.1100111111000010101101011010010 × 2⁻²⁶⁷. Rounded to 23 bits: 11001111110000101011011.

06 Subnormal result

Biased exponent -140 <= 0 -> result is subnormal (denormalized), shifted right by 140 bit(s).

d) Decimal -> Decimal (Multiplication)

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

Chat

Arithmetic

Add or multiply two operands, bit by bit.

OPERAND A

☒ Decimal

☐ Hex

12345

OPERAND B

☒ Decimal

☐ Hex

54321

OPERATION

☐ Addition (A + B)

☒ Multiplication (A × B)

Compute

12345.0 × 54321.0 = 670592768.0

SIGN 0 1 bit	EXPONENT 10011100 8 bits	MANTISSA 00111111110000110111100 23 bits
BINARY0 10011100 00111111110000110111100		

BINARY0 10011100 00111111110000110111100

HEX0x4E1FE1BC

DECIMAL670592768.0

Step-by-step

01

Convert operand A to single-precision format
A = 12345.0 -> sign=0, significand=1.10000001110010000000000 x 2¹³

02

Convert operand B to single-precision format
B = 54321.0 -> sign=0, significand=1.10101000011000100000000 x 2¹⁵

03

Multiply significands, add exponents
Significands: 1.10000001110010000000000 x 1.10101000011000100000000. Exponents: 13 + 15 = 28.
Result sign = 0 XOR 0 = 0.

04

Normalize
Raw product 100111111110000110111010010000000000000000 -> Leading 1 at 2²⁹.

05

Round (ties-to-even)
Unrounded mantissa: 1.0011111111000011011101101001000000000000000 x 2²⁹. Rounded to 23 bits: 00111111110000110111100.

06

Assemble result
Sign=0, biased exponent=156 (10011100), mantissa=00111111110000110111100.

e) Decimal -> Hex (Addition)

[illegible]

f) Decimal -> Hex (Multiplication)

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

Chat

Arithmetic

Add or multiply two operands, bit by bit.

OPERAND A

☒ Decimal
☐ Hex

12345

OPERAND B

☐ Decimal
☒ Hex

FACE

OPERATION

☐ Addition (A + B)
☒ Multiplication (A × B)

Compute

12345.0 × 8.99717692004392e-41 = 1.110701448740468e-36

SIGN	EXPONENT	MANTISSA
0 1 bit	00000111 8 bits	0111100111110011101111 23 bits

0 00000111 0111100111110011101111

BINARY

0 00000111 0111100111110011101111

HEX

0x03BCF9DF

DECIMAL

1.110701448740468e-36

Step-by-step

- Convert operand A to single-precision format**
A = 12345.0 -> sign=0, significand=1.100000111001000000000 x 2¹³
- Convert operand B to single-precision format**
B = 8.99717692004392e-41 -> sign=0, significand=0.000000111101011001110 x 2⁻¹²⁶ (subnormal)
- Multiply significands, add exponents**
Significands: 1.100000111001000000000 x 0.000000111101011001110. Exponents: 13 + -126 = -113. Result sign = 0 XOR 0 = 0.
- Normalize**
Raw product 101111001111100111011111011110000000000 -> leading 1 at 2⁻¹²⁰.
- Round (ties-to-even)**
Unrounded mantissa: 1.011110011111001110111110111110000000000 x 2⁻¹²⁰. Rounded to 23 bits: 01111001111100111011111.
- Assemble result**
Sign=0, biased exponent=7 (0000111), mantissa=0111100111110011101111.

g) Hex -> Decimal (Addition)

[illegible]

h) Hex -> Decimal (Multiplication)

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

OPERAND A

Decimal

Hex

FACE

OPERAND B

Decimal

Hex

12345

OPERATION

Addition (A + B)

Multiplication (A × B)

Compute

8.99717692004392e-41 × 12345.0 = 1.110701448740468e-36

SIGN	EXPONENT	MANTISSA
0	00000111	0111100111110011101111
1 bit	8 bits	23 bits

BINARY

0 00000111 0111100111110011101111

HEX

0x03BCF9DF

DECIMAL

1.110701448740468e-36

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

DECIMAL

1.110701448740468e-36

Step-by-step

01 Convert operand A to single-precision format

A = 8.99717692004392e-41 -> sign=0, significand=0.00000011111011001110 x 2⁻¹²⁶ (subnormal)

02 Convert operand B to single-precision format

B = 12345.0 -> sign=0, significand=1.1000001110010000000000 x 2¹³

03 Multiply significands, add exponents

Significands: 000000011111011001110 x 1100000111001000000000. Exponents: -126 + 13 = -113. Result sign = 0 XOR 0 = 0.

04 Normalize

Raw product 1011110011111001110111110000000000 -> leading 1 at 2⁻¹²⁰.

05 Round (ties-to-even)

Unrounded mantissa: 1.01111001111100111011111011110000000000 x 2⁻¹²⁰. Rounded to 23 bits: 0111100111110011101111.

06 Assemble result

Sign=0, biased exponent=7 (00000111), mantissa=0111100111110011101111.

sign · exponent · mantissa

i) Negative numbers

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

Chat

Add or multiply two operands, bit by bit.

OPERAND A

Decimal

Hex

-12345

OPERAND B

Decimal

Hex

-1234321

OPERATION

Addition (A + B)

Multiplication (A × B)

Compute

-12345.0 + -1234321.0 = -1246666.0

SIGN	EXPONENT	MANTISSA
1	10010011	00110000010111001010000
1 bit	8 bits	23 bits

BINARY

1 10010011 00110000010111001010000

HEX

0xC9982E50

HEX

0xC9982E50

DECIMAL

-1246666.0

Step-by-step

01 Convert operand A to single-precision format

A = -12345.0 -> sign=1, significand=1.10000001110010000000000 x 2¹³

02 Convert operand B to single-precision format

B = -1234321.0 -> sign=1, significand=1.001011010110010001000 x 2²⁰

03 Align exponents

Common exponent reference: 2¹³. A shifted left 0 bit(s), B shifted left 7 bit(s) to express both significands at the same scale.

04 Perform addition

Same sign -> add magnitudes: 110000001110010000000000 + 10010110101100100010000000000.

05 Normalize

Raw result 100110000010111001010000000000 -> leading 1 at 2²⁰.

06 Round (ties-to-even)

Unrounded mantissa: 1.00110000010111001010000000000 x 2²⁰. Rounded to 23 bits: 00110000010111001010000.

07 Assemble result

Sign=1, biased exponent=147 (10010011), mantissa=00110000010111001010000.

j) Infinity

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

Chat

OPERAND A

Decimal

Hex

-12345

OPERAND B

Decimal

Hex

Infinity

OPERATION

Addition (A + B)

Multiplication (A × B)

Compute

-12345.0 + Infinity = Infinity

SIGN 0 1 bit	EXPONENT 11111111 8 bits	MANTISSA 00000000000000000000000 23 bits
BINARY0 11111111 00000000000000000000000		
HEX0x7F800000		
DECIMALInfinity		

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

Chat

SIGN 0 1 bit	EXPONENT 11111111 8 bits	MANTISSA 00000000000000000000000 23 bits
BINARY0 11111111 00000000000000000000000		
HEX0x7F800000		
DECIMALInfinity		

Step-by-step

01

Convert operand A to single-precision format

A = -12345.0 -> sign=1, significand=1.100000111001000000000 x 2¹³

02

Convert operand B to single-precision format

B = Infinity -> sign=0, significand=1.00000000000000000000000 x 2¹²⁸

03

Special case: Infinity operand

B is Infinity -> result is Infinity (Infinity absorbs any finite value).

sign · exponent · mantissa

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

Chat

OPERAND A

☒ Decimal
☐ Hex

-12345

OPERAND B

☐ Decimal
☒ Hex

7F800000

OPERATION

☒ Addition (A + B)
☐ Multiplication (A × B)

Compute

-12345.0 + Infinity = Infinity

SIGN 0 1 bit	EXPONENT 11111111 8 bits	MANTISSA 000000000000000000000000 23 bits
---	---	--

BINARY

0 11111111 000000000000000000000000

HEX

0x7F800000

DECIMAL

Infinity

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

Chat

-12345.0 + Infinity = Infinity

SIGN 0 1 bit	EXPONENT 11111111 8 bits	MANTISSA 000000000000000000000000 23 bits
---	---	--

BINARY

0 11111111 000000000000000000000000

HEX

0x7F800000

DECIMAL

Infinity

Step-by-step

01 Convert operand A to single-precision format

A = -12345.0 -> sign=1, significand=1.10000001110010000000000 x 2¹³

02 Convert operand B to single-precision format

B = Infinity -> sign=0, significand=1.00000000000000000000000 x 2¹²⁸

03 Special case: Infinity operand

B is Infinity -> result is Infinity (Infinity absorbs any finite value).

sign · exponent · mantissa

k) NaN

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

OPERAND A

☒ Decimal
☐ Hex

-12345

OPERAND B

☒ Decimal
☐ Hex

NaN

OPERATION

☒ Addition (A + B)
☐ Multiplication (A × B)

Compute

-12345.0 + NaN = NaN

SIGN 0 1 bit	EXPONENT 11111111 8 bits	MANTISSA 10000000000000000000000 23 bits
---	---	---

BINARY

0 11111111 10000000000000000000000

HEX

0x7FC00000

DECIMAL

NaN

-12345.0 + NaN = NaN

SIGN 0 1 bit	EXPONENT 11111111 8 bits	MANTISSA 10000000000000000000000 23 bits
---	---	---

BINARY

0 11111111 10000000000000000000000

HEX

0x7FC00000

DECIMAL

NaN

Step-by-step

01 Convert operand A to single-precision format

A = -12345.0 -> sign=1, significand=1.1000000110010000000000 x 2¹³

02 Convert operand B to single-precision format

B = NaN -> sign=0, significand=1.10000000000000000000000 x 2¹²⁸

03 Special case: NaN

An operand is NaN -> result is NaN (invalid operation propagates).

sign · exponent · mantissa

I) Non-digit Characters

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

8x82

Arithmetic

Add or multiply two operands, bit by bit.

OPERAND A

☒ Decimal ☐ Hex

1234

OPERAND B

☒ Decimal ☐ Hex

%%\$%%3

OPERATION

☒ Addition (A + B) ☐ Multiplication (A × B)

Compute

%%\$%%3" is not a valid decimal number.

sign · exponent · mantissa

m) Type Mismatch

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

8x82

Arithmetic

Add or multiply two operands, bit by bit.

OPERAND A

☒ Decimal ☐ Hex

1234

OPERAND B

☐ Decimal ☒ Hex

hello

OPERATION

☒ Addition (A + B) ☐ Multiplication (A × B)

Compute

"hello" is not a valid hexadecimal number.

sign · exponent · mantissa

Arithmetic - IEEE 754 Machine

127.0.0.1:5000/arithmetic

Chat

8x82

Arithmetic

Add or multiply two operands, bit by bit.

OPERAND A

☒ Decimal

☐ Hex

1234

OPERAND B

☒ Decimal

☐ Hex

hello

OPERATION

☒ Addition (A + B)

☐ Multiplication (A × B)

Compute

"hello" is not a valid decimal number.

sign · exponent · mantissa